

Linux Kernel Architecture

User Space

Library Interface

Kernel Space

Scheduler

Manages CPU resource allocation and process scheduling

Memory Management

Handles memory allocation, paging, and virtual memory

File System

Provides interfaces for file operations and manages data storage

Network Stack

Implements network protocols and facilitates network communication

Short Descriptions of Linux Kernel Components

1. Process Scheduler

The scheduler manages process execution by determining which process runs at any given time. It ensures fairness and efficiency, supporting various scheduling algorithms for multitasking and time-sharing.

2. Memory Management Unit (MMU)

This component handles the allocation and deallocation of memory to processes. It supports virtual memory, page swapping, and memory protection to ensure stability and performance.

3. File System

The file system manages how data is stored, organized, and retrieved on storage devices. It supports various formats like ext4, FAT, and NTFS, and provides abstraction for file operations via system calls.

4. Network Stack

The network stack enables communication over networks using standard protocols like TCP/IP. It handles packet routing, error checking, and ensures reliable data transmission between devices.

5. Inter-Process Communication (IPC)

IPC mechanisms like pipes, shared memory, message queues, and semaphores allow processes to communicate and synchronize their actions. This is crucial for multitasking and complex application execution.

6. Device Drivers

Device drivers act as translators between the hardware and the kernel. Each driver provides a standard interface for the OS to interact with hardware devices such as printers, disks, or USBs.

7. System Call Interface (SCI)

The SCI provides a bridge for user-space applications to interact with kernel functions. It translates user requests into kernel-level instructions for resource access and control.

8. Kernel Mode and User Mode

The Linux kernel operates in kernel mode with unrestricted access to all system resources, whereas user applications run in user mode with limited access. This separation enhances system security and stability.